Mathematical Questions, Equations, and Symbol Meanings

This section presents common questions relevant to portfolio analysis, Monte Carlo simulation, and GARCH validation. Each equation is followed by definitions of its symbols and an interpretation. Some equations are general financial reference formulas; whether a particular metric is calculated by a script depends on that script's implementation.

Question 1. How is daily log return calculated?

Equation

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Symbols

Symbol	Meaning
r_t	Log return at time t, dimensionless
P_t	Asset price at time t
P_{t-1}	Asset price at the preceding observation
$\ln(\cdot)$	Natural logarithm
t	Current observation or time index

Interpretation: Log return measures the continuously compounded change between two positive prices. It is useful for statistical modelling and is the return representation used by the simulation workflow.

Question 2. How is simple return calculated, and how is it related to log return?

Equations

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$r_t = \ln(1 + R_t), \qquad R_t = e^{r_t} - 1$$

Symbols

Symbol	Meaning
R_t	Simple return over the period
r_t	Log return over the period
P_t	Current price
P_{t-1}	Previous price
e	Euler's number, the base of natural logarithms

Interpretation: A simple return of 0.05 represents a 5% increase. The log and simple returns are related but are not interchangeable in portfolio aggregation.

Question 3. How is the historical mean return calculated?

Equation

$$\bar{r} = \frac{1}{T} \sum_{t=1}^{T} r_t$$

Symbols

Symbol	Meaning
$\bar{r}$	Sample mean return
r_t	Return observed at time t
T	Number of return observations
$\sum$	Summation over the specified observations
t	Observation index

Interpretation: The mean summarizes the average return per observation in the selected historical sample. It is not necessarily the expected return that will occur in the future.

Question 4. How are sample variance and standard deviation calculated?

Equations

$$s^2 = \frac{1}{T-1} \sum_{t=1}^{T} (r_t - \bar{r})^2$$

$$s = \sqrt{s^2}$$

Symbols

Symbol	Meaning
s^2	Sample variance
s	Sample standard deviation
r_t	Return at observation t
$\bar{r}$	Sample mean return
T	Number of observations
$T-1$	Degrees-of-freedom adjustment for sample variance
$\sqrt{\cdot}$	Square-root operator

Interpretation: Variance measures dispersion in squared-return units; standard deviation expresses dispersion in return units. Daily volatility should not be confused with annualized volatility.

Question 5. How are covariance and correlation between two assets calculated?

Equations

$$\mathrm{Cov}(r_i, r_j) = \frac{1}{T-1} \sum_{t=1}^{T} (r_{i,t} - \bar{r}_i)\,(r_{j,t} - \bar{r}_j)$$

$$\rho_{ij} = \frac{\mathrm{Cov}(r_i, r_j)}{s_i\, s_j}$$

Symbols

Symbol	Meaning
$\mathrm{Cov}(r_i, r_j)$	Sample covariance between assets i and j
ρ_{ij}	Correlation coefficient between assets i and j
$r_{i,t}$	Return of asset i at time t
$r_{j,t}$	Return of asset j at time t
$\bar{r}_i,\ \bar{r}_j$	Sample mean returns for assets i and j
$s_i,\ s_j$	Sample standard deviations for assets i and j
i, j	Asset indices
T	Number of aligned observations

Interpretation: Covariance captures how two return series move together in their original units. Correlation standardizes that relationship and lies between -1 and $+1$, when both series have nonzero variance.

Question 6. How is a portfolio's simple return calculated from asset simple returns?

Equation

$$R_{p,t} = \sum_{i=1}^{N} w_i \, R_{i,t}$$

with

$$\sum_{i=1}^{N} w_i = 1$$

Symbols

Symbol	Meaning
$R_{p,t}$	Portfolio simple return at time t
$R_{i,t}$	Simple return of asset i at time t
w_i	Portfolio weight assigned to asset i
N	Number of assets in the portfolio
$\sum_i w_i = 1$	Fully invested portfolio-weight constraint

Interpretation: For a fully invested portfolio with weights maintained over the return period, the portfolio simple return is the weighted sum of asset simple returns. The relationship must be applied carefully when the simulated inputs are log returns.

Question 7. How is portfolio variance calculated from a covariance matrix?

Equations

$$\sigma_p^2 = \mathbf{w}^\mathsf{T} \, \mathbf{\Sigma} \, \mathbf{w}$$

$$\sigma_p = \sqrt{\mathbf{w}^\mathsf{T} \, \mathbf{\Sigma} \, \mathbf{w}}$$

Symbols

Symbol	Meaning
σ_p^2	Portfolio variance
σ_p	Portfolio standard deviation
$\mathbf{w}$	Column vector of portfolio weights
$\mathbf{w}^\mathsf{T}$	Transpose of the weight vector
$\mathbf{\Sigma}$	Asset-return covariance matrix
N	Number of assets represented in the vector and matrix

Interpretation: Portfolio risk depends on asset weights and the full covariance matrix, not only on each asset's individual volatility. Covariance is the mathematical basis for modelling cross-asset dependence in the multivariate-normal simulation.

Question 8. What distribution does the multivariate-normal model assume for asset log returns?

Equation

$$\mathbf{r}_t \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Symbols

Symbol	Meaning
$\mathbf{r}_t$	Vector of asset log returns at time t
$\sim$	"Is distributed according to"
$\mathcal{N}$	Multivariate normal distribution
$\boldsymbol{\mu}$	Vector of estimated mean log returns
$\boldsymbol{\Sigma}$	Covariance matrix of log returns
t	Simulated time index

Interpretation: This model represents dependence through a covariance matrix. Its standard form does not capture all features that can occur in financial returns, such as heavy tails, changing correlations, or volatility clustering.

Question 9. How is portfolio value updated when the simulated inputs are log returns and the portfolio is rebalanced to target weights?

For a portfolio rebalanced to weights w_i at each period, the simple portfolio return implied by asset log returns is

$$R_{p,t+1} = \sum_{i=1}^{N} w_i \left(e^{r_{i,t+1}} - 1\right)$$

and the corresponding value update is

$$V_{t+1} = V_t \left[1 + \sum_{i=1}^{N} w_i \left(e^{r_{i,t+1}} - 1\right)\right].$$

Symbols

Symbol	Meaning
V_t	Portfolio value at time t
V_{t+1}	Portfolio value at the next period
$r_{i,t+1}$	Simulated log return for asset i over the next period
$e^{r_{i,t+1}} - 1$	Simple return corresponding to the simulated log return
w_i	Target portfolio weight for asset i
N	Number of portfolio assets

Interpretation: This equation is appropriate for a daily-rebalanced portfolio with fixed target weights and no fees, taxes, cash flows, or market frictions. Verify the value-update expression in the specific simulation script before treating this as an exact description of its implementation.

Question 10. How is compound annual growth rate (CAGR) calculated?

Equation

$$\text{CAGR} = \left(\frac{V_{\text{end}}}{V_{\text{start}}}\right)^{1/Y} - 1$$

Symbols

Symbol	Meaning
V_{end}	Portfolio value at the end of the period
V_{start}	Portfolio value at the start of the period
Y	Length of the period in years
CAGR	Compound annual growth rate

Interpretation: CAGR expresses the constant annual growth rate that connects the starting and ending values over the specified period. This is a reference formula; do not assume the Monte Carlo script reports CAGR unless the relevant output confirms it.

Question 11. How is annualized volatility estimated from daily volatility?

Equation

$$\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{D}$$

Symbols

Symbol	Meaning
σ_{annual}	Annualized standard deviation
σ_{daily}	Daily standard deviation
D	Assumed number of trading days per year; commonly 252
$\sqrt{D}$	Square root of the annualization factor

Interpretation: This is a conventional annualization under assumptions about the return process. It is not a forecast that volatility will remain constant throughout the year.

Question 12. How is the Sharpe ratio calculated?

Equation

$$\text{Sharpe} = \frac{\bar{R}_p - R_f}{s_p}$$

Symbols

Symbol	Meaning
$\bar{R}_p$	Average portfolio return per period
R_f	Risk-free return for the same period and in compatible units
s_p	Standard deviation of portfolio returns for the same period
Sharpe	Return in excess of the risk-free rate per unit of return standard deviation

Interpretation: The Sharpe ratio compares excess return with variability. If the numerator uses daily returns and the denominator uses daily standard deviation, the result is a daily-period ratio; annualization requires consistent treatment of both quantities. This is a reference formula and should only be reported as a project result if it is calculated by the relevant script.

Question 13. What is Value at Risk (VaR)?

Let L denote portfolio loss under a clearly specified loss convention. The loss quantile at confidence level α is

$$\text{VaR}_\alpha(L) = F_L^{-1}(\alpha)$$

where F_L is the cumulative distribution function of loss.

Symbols

Symbol	Meaning
L	Portfolio loss, defined according to the report's convention
$\mathrm{VaR}_\alpha(L)$	Loss quantile at confidence level α
α	Quantile level, such as 0.95 or 0.99
F_L	Cumulative distribution function of loss
F_L^{-1}	Quantile function, or inverse CDF under the chosen convention

Interpretation: At a 95% loss quantile, approximately 95% of modelled losses are at or below the VaR threshold, with approximately 5% above it, subject to ties and the quantile convention. VaR does not describe the average size of losses beyond the threshold. The report must state whether loss is defined relative to starting value, a benchmark, or another reference.

Question 14. What is Expected Shortfall (ES)?

For a continuous loss distribution, Expected Shortfall at level α can be written as

$$\mathrm{ES}_\alpha(L) = \mathbb{E}[\, L \mid L \geq \mathrm{VaR}_\alpha(L) \,] \,.$$

Symbols

Symbol	Meaning
$\mathrm{ES}_\alpha(L)$	Expected Shortfall at level α
$\mathbb{E}[\cdot]$	Expected value or mean
L	Portfolio loss
$\mathrm{VaR}_\alpha(L)$	VaR threshold
α	Confidence level
$\mid$	"Conditional on"
$\geq$	Greater than or equal to

Interpretation: ES estimates the average loss in the tail beyond the VaR threshold. For finite Monte Carlo samples or discrete distributions, the exact empirical calculation depends on how tail observations and quantiles are selected; consult the implementation for its convention.

Question 15. What is a GARCH(1,1) model?

Equations

$$r_t = \mu + \varepsilon_t$$

$$\varepsilon_t = \sigma_t \, z_t$$

$$\sigma_t^2 = \omega + \alpha_1 \, \varepsilon_{t-1}^2 + \beta_1 \, \sigma_{t-1}^2$$

A common sufficient parameter condition for a finite unconditional variance in the standard model is

$$\omega > 0, \qquad \alpha_1 \geq 0, \qquad \beta_1 \geq 0, \qquad \alpha_1 + \beta_1 < 1.$$

Symbols

Symbol	Meaning
r_t	Return at time t
μ	Conditional mean or mean parameter, according to the specification
ε_t	Return innovation or residual at time t
σ_t	Conditional standard deviation at time t
σ_t^2	Conditional variance at time t
z_t	Standardized innovation, commonly with mean zero and variance one
ω	Variance-equation constant
α_1	Coefficient on the previous squared innovation (ARCH term)
β_1	Coefficient on the previous conditional variance (GARCH term)
$t-1$	Previous observation
$(1,1)$	One lag of squared innovations and one lag of conditional variance

Interpretation: GARCH models volatility as changing over time. Large recent shocks can raise conditional variance, and high variance can persist. The parameter condition shown is a standard stationarity condition for the usual GARCH(1,1) specification, not a guarantee that the model fits every return series well.

Question 16. What is standardized-residual bootstrap sampling, and why resample residual vectors jointly?

For asset i, a standardized residual is

$$z_{i,t} = \frac{\varepsilon_{i,t}}{\sigma_{i,t}}.$$

If residuals from all N assets are aligned by date, a sampled residual vector can be represented as

$$\mathbf{z}_t^* = \begin{bmatrix} z_{1,k} \\ z_{2,k} \\ \vdots \\ z_{N,k} \end{bmatrix},$$

where k is a sampled historical index shared by all assets in that vector.

Symbols

Symbol	Meaning
$z_{i,t}$	Standardized residual of asset i at time t
$\varepsilon_{i,t}$	Estimated innovation for asset i
$\sigma_{i,t}$	Estimated conditional standard deviation for asset i
$\mathbf{z}_t^*$	Resampled vector of standardized residuals for a simulated period
N	Number of assets
k	Historical row selected for resampling
$*$	Indicates a bootstrap-sampled quantity

Interpretation: Sampling the same historical row across assets preserves the observed cross-sectional pattern of standardized residuals at that date. It does not guarantee that every dependence feature of future markets is reproduced.

Question 17. How is QLIKE loss calculated for variance forecasts?

Equation

$$\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left[\ln(h_t) + \frac{y_t}{h_t}\right]$$

Symbols

Symbol	Meaning
QLIKE	Average quasi-likelihood loss
T	Number of evaluated holdout observations
h_t	Positive forecast variance for observation t
y_t	Nonnegative realized-variance proxy for observation t, such as squared return
$\ln(h_t)$	Natural logarithm of the forecast variance

Symbol	Meaning
$\sum$	Sum across the evaluation observations

Interpretation: Lower QLIKE is generally preferred when models are evaluated on the same observations with the same realized proxy and compatible conventions. In this project, squared returns are used as a proxy for realized variance, so the score should not be interpreted as comparison against perfectly observed latent volatility.

Implementation checks: Forecast variances must be strictly positive; realized proxies must be nonnegative; and the forecast and realized arrays must be aligned and of equal length. The project tests invalid inputs and the chronological split.

Question 18. How can the GARCH and historical-variance forecasts be compared fairly?

A useful comparison is the average QLIKE score for each model over the same holdout dates and the same realized-variance proxy:

$$\Delta_{\text{QLIKE}} = \text{QLIKE}_{\text{historical}} - \text{QLIKE}_{\text{GARCH}} .$$

Symbols

Symbol	Meaning
Δ_{QLIKE}	Difference between the two models' QLIKE scores
$\text{QLIKE}_{\text{historical}}$	QLIKE of the expanding historical-variance baseline
$\text{QLIKE}_{\text{GARCH}}$	QLIKE of the GARCH model
Positive Δ_{QLIKE}	GARCH has the lower QLIKE score in this defined comparison
Negative Δ_{QLIKE}	Historical-variance baseline has the lower QLIKE score

Interpretation: This sign convention makes a positive difference favor GARCH. The actual output files must be inspected to determine which model performed better; do not infer model superiority from the model names alone.

General mathematical notation glossary

Symbol or notation	Meaning
t	Time or observation index
i, j	Asset indices

Symbol or notation	Meaning
N	Number of assets
T	Number of observations
$\bar{x}$	Sample mean of x
s^2	Sample variance
s, σ	Standard deviation; context determines whether it is sample or conditional
Σ	Covariance matrix
ρ	Correlation coefficient
μ	Mean parameter or vector
$\omega, \alpha_1, \beta_1$	GARCH variance parameters
$\sum$	Summation
ln	Natural logarithm
$\exp(x)$ or e^x	Exponential function
$\sqrt{x}$	Nonnegative square root
x^2	x multiplied by itself
$\mathbb{E}[X]$	Expected value of random variable X
$\sim$	Follows a stated probability distribution
$\geq$, $\leq$	Greater than or equal to; less than or equal to
T	Matrix or vector transpose
$\mathbf{x}$	A vector
$\mathbf{\Sigma}$	A matrix
$1 - \alpha$	Tail probability associated with quantile level α
$*$	Bootstrap-sampled value when used as a superscript in this section